TEST 1 INNOVATOR

MATH MODULE 1

1. The values below represent how many points Posh scored in each of his last ten basketball games.

$$22, 14, 15, 8, 24, 14, 14, 17, 22, 30$$

What is the mean of Posh's scores over these ten games?

2. A lemonade stand sells small and large cups. Small cups are sold for $\$2.25$, and large cups are sold for $\$3.50$. The stand made $\$31.50$ selling large cups. If the stand sold the same number of small cups as large cups, what was the total amount of money, in dollars, that the stand made?

PREV. QUESTION NEXT QUESTION

× A 20.25

× B 47.25

× C 51.75

× D 63.00

3. In triangle BMD, the measure of angle M is 38° and the measure of the exterior angle of angle D is 114°. What is the measure of angle B, in degrees?

PREV. QUESTION NEXT QUESTION

× A 28

× B 62

× C 76

× D 152

4. Of the $52{,}000$ seats in a baseball stadium, $44{,}200$ are in foul territory. Out of those seats in foul territory, $17{,}680$ are painted blue. What percentage of the seats in the baseball stadium are in foul territory and not painted blue?

PREV. QUESTION | NEXT QUESTION

- A 40%
- B 51%
- C 60%
- D 66%

5. $$0.1a - 0.1b = 1.2$$
$$\frac{1}{5}a - \frac{1}{2}b = \frac{3}{10}$$

Which of the following ordered pairs (a, b) satisfies the given system of equations?

PREV. QUESTION | NEXT QUESTION

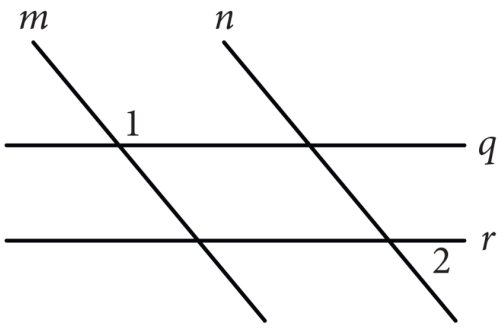
- A $(5, 7)$
- B $(7, -5)$
- C $(7, 19)$
- D $(19, 7)$

6. Ryan started a tutoring company. The total amount paid by a client monthly, in dollars, is represented by the linear function $f(x) = 24.45x + 10.50$, where x is the number of tutoring sessions given to the client per month. What is the best interpretation of the value 24.45 in the function?

PREV. QUESTION | NEXT QUESTION

- A Ryan charges $\$24.45$ per month.
- B Ryan charges $\$24.45$ per session.
- C Clients pay an initial fee of $\$24.45$.
- D Clients pay a monthly fee of $\$24.45$.

7.



In the figure given, lines q and r are parallel, and lines m and n are parallel. If the measure of angle 1 is 130°, what is the measure of angle 2?

PREV. QUESTION | NEXT QUESTION

- A 40°
- B 50°
- C 65°
- D 130°

COME BACK LATER

8. If $4x + 16 = 24$, what is the value of $x + 4$?

PREV. QUESTION | NEXT QUESTION

- A 2
- B 4
- C 6
- D 8

9. Which of the following is equivalent to $(3a^2 + 6b^2)^2$?

PREV. QUESTION | NEXT QUESTION

- A $3a^4 + 18a^2b^2 + 36b^4$
- B $3a^2 + 36a^2b^2 + 36b^4$
- C $9a^4 + 36a^2b^2 + 36b^4$
- D $9a^4 + 18a^2b^2 + 36b^4$

10. $\frac{\sqrt{x^2-9}}{a} = 1$

In the given equation, $x < 0$ and $a = 4$. What is the value of $x - 5$?

PREV. QUESTION | NEXT QUESTION

A -13

B -10

C -5

D -3

11. In the xy-coordinate plane, the point $(3, 8)$ lies on the graph of the function $f(x) = x^3 - bx^2 - x + 2$, and b represents a constant. What is the value of b?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

12. In May 2022, $1{,}200$ people were randomly polled in a district with an expected voter turnout of $80{,}000$. The poll estimated that 45% of the voters in the district who turn out on election day would vote for the independent candidate, with an associated margin of error of 3.5%. Which of the following is a likely value for the total number of voters in that district who vote for the independent candidate?

PREV. QUESTION | NEXT QUESTION

A 540

B $2{,}800$

C $35{,}450$

D $48{,}500$

13. $4x^4 + 20x^2y + 25y^2$

Which of the following is equivalent to the given expression?

PREV. QUESTION | NEXT QUESTION

× A $(4x^2 + 5y^2)^2$

× B $(4x^2 + 10y^2)^2$

× C $(2x^2 + 5y^2)^2$

× D $(2x^2 + 5y)^2$

14. $$ax + by = 16$$
$$3x + 2y = 64$$

In the given system of equations, a and b are constants. If the system has infinitely many solutions, what is the value of ab?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

15. Iodine-131 is a radioisotope used in medicine. It has a half-life of about 8 days, which means it takes about 8 days for a sample to decay to half of its original amount. If a patient takes a dose that contains 100mCi of Iodine-131, which of the following functions best expresses the amount of y, in mCi, that remains in that patient's system t days after taking that dose?

PREV. QUESTION | NEXT QUESTION

× A $y = \frac{1}{2}(100)^{8t}$

× B $y = 100(\frac{1}{2})^{8t}$

× C $y = 100(\frac{1}{2})^{\frac{t}{8}}$

× D $y = \frac{1}{2}(100)^{\frac{t}{8}}$

16. In triangle PQR, $\sin(P) = \frac{7}{25}$, and angle R is a right angle. What is the value of $\sin(Q)$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

17. $$2(12x - 5) = 3(mx - 4)$$

In the given equation, m is a constant. If the equation has no solution, what is the value of m?

PREV. QUESTION | NEXT QUESTION

A 4

B 8

C 15

D 18

18. $$\frac{11}{15}x - \frac{8}{15}x = \frac{1}{6} + \frac{1}{12}$$

What value of x satisfies the given equation?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

19. Jupiter travels a distance of approximately 4.9 billion (or 10^9) kilometers during its orbit around the sun. Jupiter completes an orbit in about 12 Earth years. Which of the following is closest to the number of kilometers Jupiter travels in one hour?

PREV. QUESTION | NEXT QUESTION

- A 47
- B $1{,}100$
- C $47{,}000$
- D $560{,}000$

20. If $(5^b)^{3a} = 25$, what is the value of $36ab$?

PREV. QUESTION | NEXT QUESTION

- A 3
- B 4
- C 12
- D 24

21. $$g(x) = 3x^2 - 16x + 48$$

The given equation defines the function g. For what value of x does $g(x)$ reach its minimum?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

22. $x^2 + y^2 - 10y = 9$

The given equation defines a circle in the xy-plane. Which of the following is the length of the circle's radius?

PREV. QUESTION | NEXT QUESTION

- A 3
- B $\sqrt{10}$
- C 5
- D $\sqrt{34}$

MATH
MODULE 2

1. If x is $2k\%$ of y, what is the value of $3k\%$ of $0.3y$?

PREV. QUESTION | NEXT QUESTION

- A $0.05x$
- B $0.2x$
- C $0.45x$
- D $0.9x$

2. $h(x) = 3x^3 - ax$

For the given function $h(x)$, $h(2) = 20$. What is the value of $h(-1)$?

PREV. QUESTION | NEXT QUESTION

- A -5
- B -1
- C 1
- D 5

3. An angle opposite a leg of a right triangle measures w°, and $\tan(w^\circ) = \frac{3}{4}$. What is $\sin(90^\circ - w^\circ)$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

4. Which of the following is equivalent to $x^{0.6} \cdot x^{\frac{3}{20}}$ for all positive values of x?

PREV. QUESTION | NEXT QUESTION

A $\sqrt[20]{x^{0.9}}$

B $\sqrt[5]{x^3}$

C $\sqrt[4]{x^3}$

D $\sqrt[3]{x^4}$

5. $$3.5a - 4.25b = -14.25$$
$$4.25a - 3.5b = -9$$

The solution to the given system of equations is (a, b). What is the value of $a - b$?

PREV. QUESTION | NEXT QUESTION

A -7

B -3

C 5

D 7

6. A survey counted the number of flowers on the bushes in a park. The mean number of flowers per bush was 34 while the median number was 45. What factor most likely would account for the difference between the mean and the median?

PREV. QUESTION | NEXT QUESTION

- A. There were some bushes with extremely few flowers.
- B. The mode of the data was a number greater than 34.
- C. Many bushes had between 34 and 45 flowers.
- D. There is little variance in the number of flowers on the bushes.

7. The table shows the distribution of coffee preferences among the 27 members of the DC Book Club.

	Cream	No Cream	Total
Sugar	4	7	11
No Sugar	15	1	16
Total	19	8	27

If a club member is selected at random, what is the probability the member prefers no cream, given that they prefer sugar?

PREV. QUESTION | NEXT QUESTION

- A. $\frac{7}{27}$
- B. $\frac{7}{16}$
- C. $\frac{7}{11}$
- D. $\frac{7}{8}$

8. A line in the xy-plane contains the point $(3, -5)$ and has a slope of $\frac{4}{3}$. Which of the following points lies on the line?

PREV. QUESTION | NEXT QUESTION

- A. $(\frac{1}{2}, 1\frac{2}{3})$
- B. $(1\frac{2}{3}, -5)$
- C. $(2, -6\frac{1}{3})$
- D. $(5, -5\frac{1}{4})$

9. The height of a cone is three times its diameter. The base of the cone is a circle, and the area of the base is 9π square centimeters. What is the volume of the cone, in cubic centimeters?

PREV. QUESTION | NEXT QUESTION

- A 27π
- B 54π
- C 162π
- D 216π

10. A formula used for compound interest is $A = P(1 + \frac{r}{n})^{nt}$. Which of the following represents P in terms of A, n and t?

PREV. QUESTION | NEXT QUESTION

- A $\frac{(1+\frac{r}{n})^{nt}}{A}$
- B $A(1 + \frac{r}{n})^{nt}$
- C $\frac{A}{(1+\frac{r}{n})^{nt}}$
- D $\frac{1}{A(1+\frac{r}{n})^{nt}}$

11. A parking garage charges $\$11$ on weekdays and $\$15$ on weekends. If the garage brought in $\$53{,}234$ in one week and a total of $4{,}122$ people used the garage, how many people used the garage during the weekend?

PREV. QUESTION | NEXT QUESTION

- A 304
- B $1{,}973$
- C $2{,}149$
- D $3{,}549$

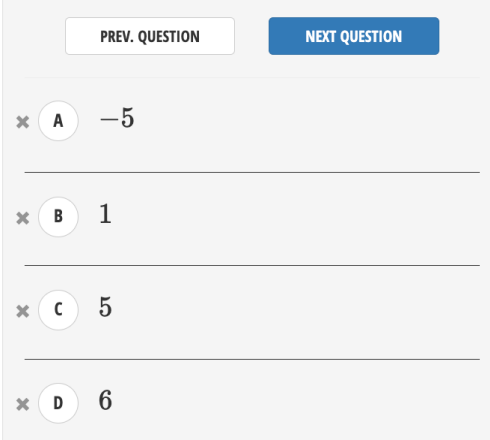
12. If $\frac{a^{2x^3}}{a^4} = a^{12}$, what is the value of 2^x?



13. If $(3x + a)(2x + b) = 6x^2 + cx + 6$ and $4a + 6b = 36$ for all values of x, what is the value of c?



14. The graph of line q is perpendicular to the line $2x - 4y = 11$ and contains the point $(1, 6)$. Line p contains the points $(2\frac{1}{4}, 7\frac{1}{4})$ and $(\frac{1}{3}, 5\frac{1}{3})$. If p and q intersect at the point (a, b), what is the value of $a - b$?



15. In triangles ABC and DEF, $\angle A$ and $\angle D$ each measure 48°, and the measures of $\overline{AB}$ and $\overline{DE}$ are both 5 inches. Which of the following is sufficient to prove that ABC and DEF are congruent triangles?

PREV. QUESTION | NEXT QUESTION

× A $\overline{BC}$ measures 3 inches and $\overline{EF}$ measures 4 inches.

× B $\overline{BC}$ measures 4 inches and $\overline{EF}$ measures 4 inches.

× C $\overline{AC}$ measures 4 inches and $\overline{DF}$ measures 4 inches.

× D $\overline{BC}$ measures 3 inches and $\overline{DF}$ measures 4 inches.

16. $$f(x) = -2x^2 + 16x - 11$$

If $g(x) = f(x - 3)$, what is the value of x where $g(x)$ reaches its maximum value?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

17. Which expression is equivalent to $\frac{25(x^2-5)}{(x-\sqrt{5})}$, where $x \neq \sqrt{5}$?

PREV. QUESTION | NEXT QUESTION

× A $5(x + 5)$

× B $5(x - 5)$

× C $25(x + \sqrt{5})$

× D $25(x - \sqrt{5})$

18. The equation $ax + b = cx + d$ relates the positive numbers a, b, c, d, and x. Which equation correctly expresses x in terms of a, b, c, and d?

PREV. QUESTION | NEXT QUESTION

A $x = a - c + d - b$

B $x = \frac{d-b}{a-c}$

C $x = \frac{d-b}{c-a}$

D $x = d - b - \frac{c}{a}$

19. Wendy spent $\$174.40$ on a bulk supply of pens. Each box of pens costs $\$25$. A shipping charge of $\$p$ is added. A 9% tax is applied to both the total order and the shipping charge. If Wendy ordered 6 boxes, what is the value of p?

PREV. QUESTION | NEXT QUESTION

A 10.00

B 10.90

C 13.50

D 41.65

20. $$f(x) = -200(3)^x$$
$$g(x) = -200(3)^{x+a}$$

In the equations given, a is a positive constant. Which of the following statements about the functions f and g must be true ?

I. The y-intercept of the graph of g is less than the y-intercept of the graph f.

II. The y-intercept of the graph of g is greater than the y-intercept of the graph of f.

III. If $a < b$, then $f(a) < f(b)$.

PREV. QUESTION | NEXT QUESTION

A I only

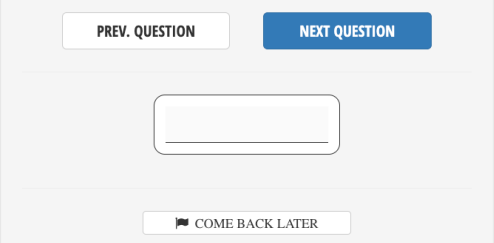
B II only

C I and III only

D II and III only

21. $$x^2 + y^2 - 6x - 8y = -16$$

The equation above defines a circle in the xy-coordinate plane. What is the length of the circle's radius?



22. A sequence of numbers starts at the number a. Each sequential number is k more than the previous number. The sum of the first 13 numbers can be written as $xa + yk$. What is the value of $x + y$?

